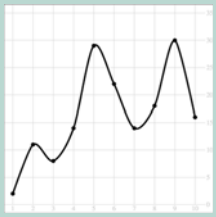
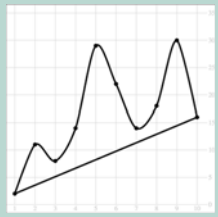
Should you always draw filled paths with d3.svg.area?

No. Counterintuitively, you should use `d3.svg.line` to draw filled areas. To do so, though, you need to append *Z* to the created `d` attribute. This indicates that the path is closed.

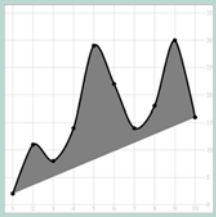
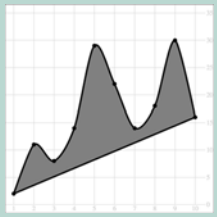
Open path	Closed path changes	Explanation

`movieArea = d3.svg.line()`
`  .x(function(d) {`
`  return xScale(d.day)`
`  })`
`  .y(function(d) {`
`  return yScale(d[x])`
`  })`
`  .interpolate("cardinal");`
		You write the constructor for the line-drawing code the same regardless of whether you want a line or shape, filled or unfilled.

`d3.select("svg")`
`  .append("path")`
`  **.attr("d", movieArea(data))**`
`  .attr("fill", "none")`
`  .attr("stroke", "black")`
`  .attr("stroke-width", 3);`
	`d3.select("svg")`
	`  .append("path")`
	`  **.attr("d", movieArea(data)`**
	`  **+ "Z")**`
	`  .attr("fill", "none")`
	`  .attr("stroke", "black")`
	`  .attr("stroke-width", 3);`
		When you call the constructor, you append a `<path>` element. You specify whether the line is "closed" by concatenating a *Z* to the string created by your line constructor for the `d` attribute of the `<path>`.

		When you add a *Z* to the end of an SVG `<path>` element's `d` attribute, it draws a line connecting the two end points.

`d3.select("svg")`
`  .append("path")`
`  .attr("d", movieArea(data))`
`  **.attr("fill", "none")**`
`  .attr("stroke", "black")`
`  .attr("stroke-width", 3);`
	`d3.select("svg")`
	`  .append("path")`
	`  .attr("d", movieArea(data)`
	`  + "Z")`
	`  **.attr("fill", "gray")**`
	`  .attr("stroke", "black")`
	`  .attr("stroke-width", 3);`
		You may think that only a closed path could be filled, but the fill of a path is the same whether or not you close the line by appending *Z*.

		The area of a path filled is always the same, whether it's closed or not.